Histology
Trachea & lungs

MANIPAL
UNIVERSITY COLLEGE
MALAYSIA

Formerly known as MELAKA-MANIPAL MEDICAL COLLEGE

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Only for instruction purpose.

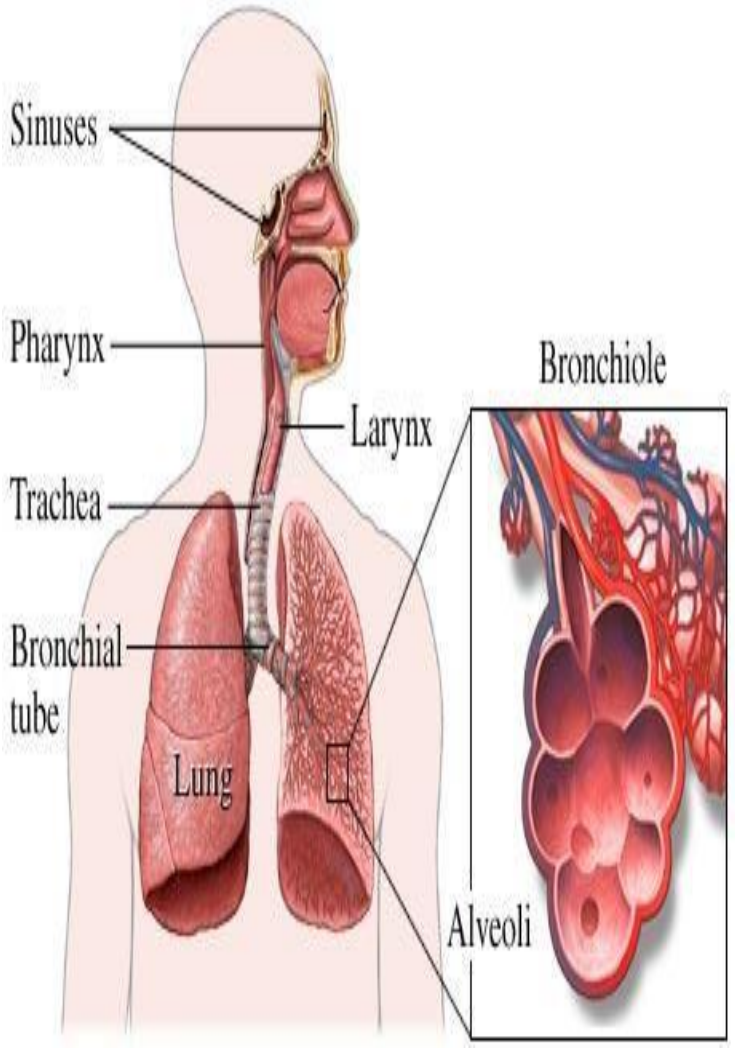
#inspire

OBJECTIVES

Explain and illustrate the microscopic structure of Trachea and Lungs.

INTRODUCTION

- The respiratory system is concerned with delivery of oxygen from the atmosphere to tissues and excretion of carbon dioxide from tissues to atmosphere.
- The respiratory system consists of nose , pharynx , larynx , trachea and lungs.



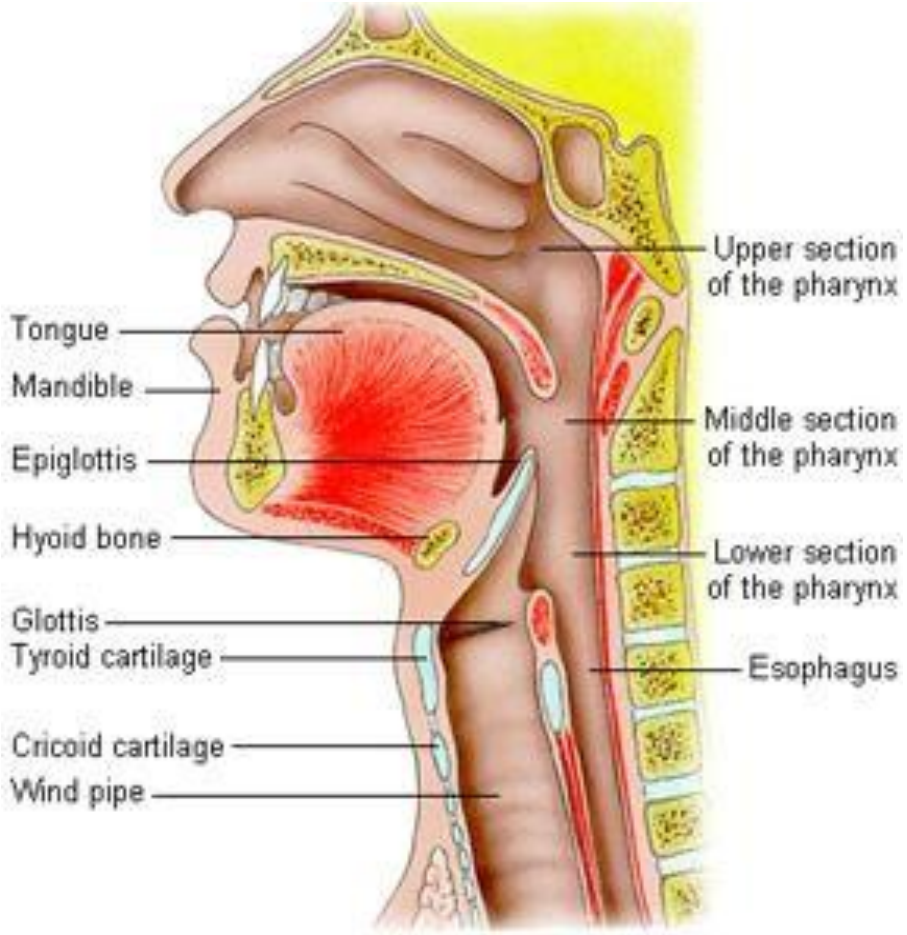
FUNCTIONAL COMPONENTS – RESPIRATORY SYSTEM

- Conducting portion

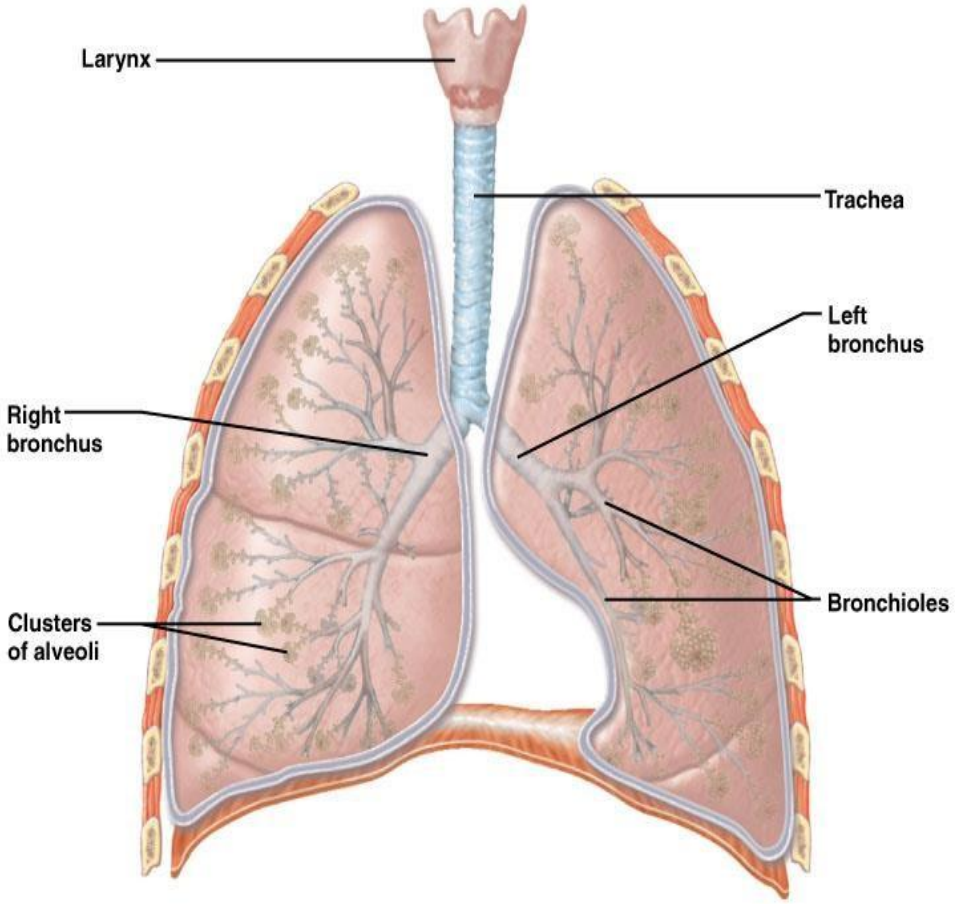
- Respiratory portion

CONDUCTING SYSTEM

Extrapulmonary part

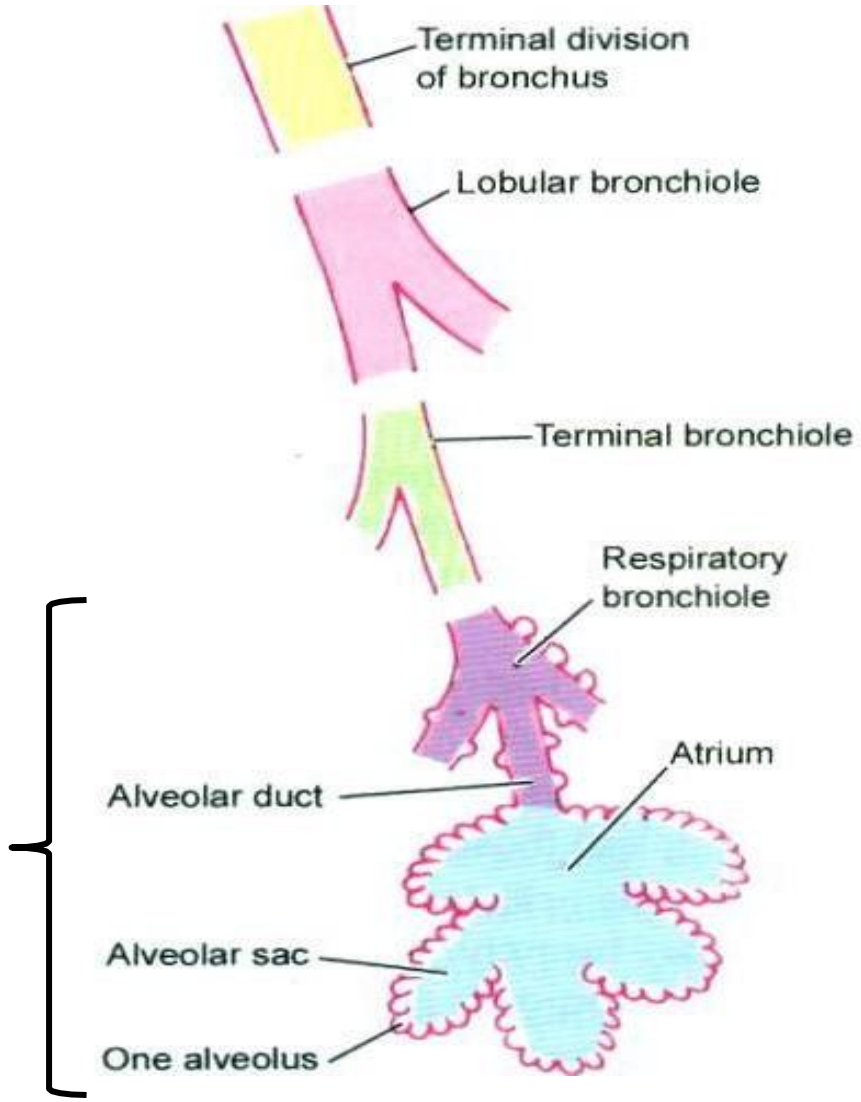


Intrapulmonary part



RESPIRATORY SYSTEM

- Involved in exchange of oxygen and carbon dioxide between blood and inspired air.
- It includes terminal parts of lower respiratory tract:-
- Respiratory bronchioles , alveolar ducts , alveolar sacs and alveoli.



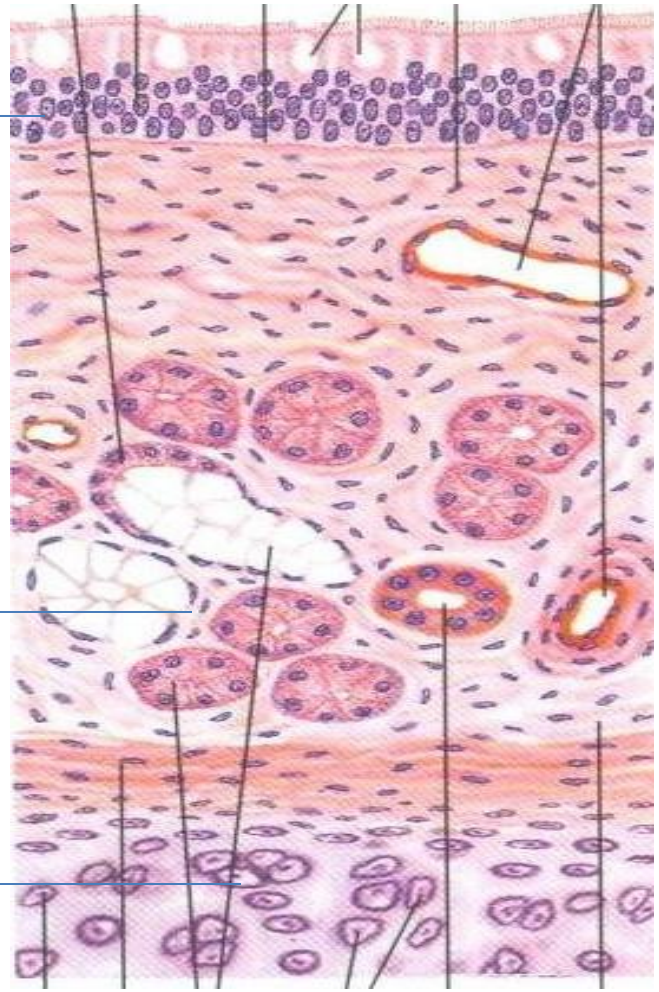
GENERAL STRUCTURE OF CONDUCTING SYSTEM

Mucosa

Submucosa

Cartilage layer

Adventia



MUCOSA

Respiratory epithelium contains following types of cells

Ciliated columnar cell

Goblet cell

Clara cell

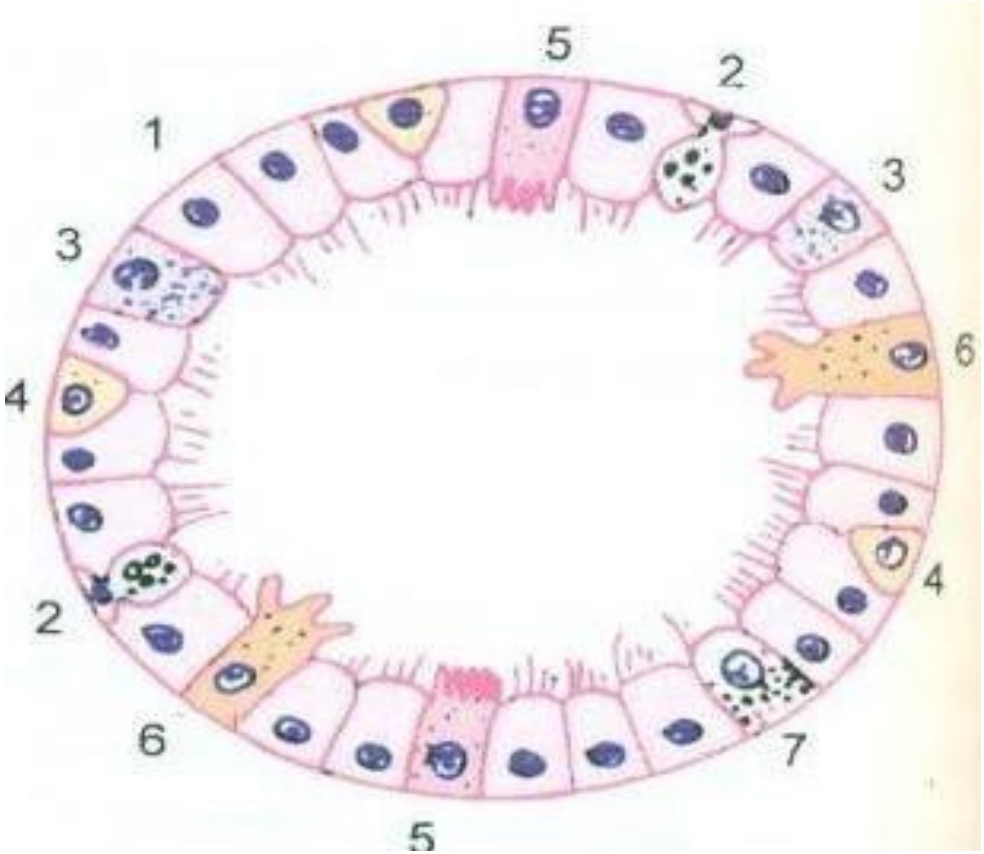
Brush cell

Neuroendocrine cell

Basal cell

Lymphocytes

Mast cell



CILIATED COLUMNAR CELLS

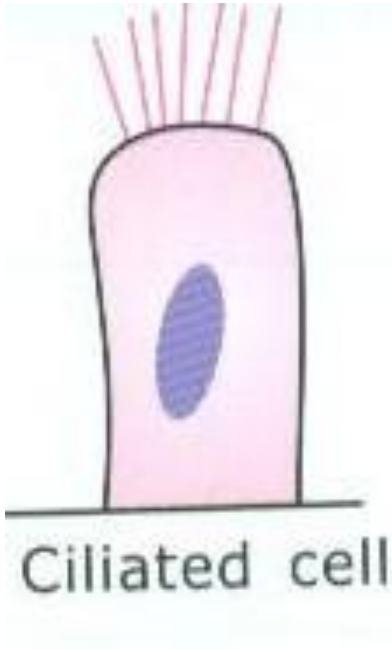
Columnar cells with cilia on their free surface

Location : extrapulmonary and intrapulmonary part.

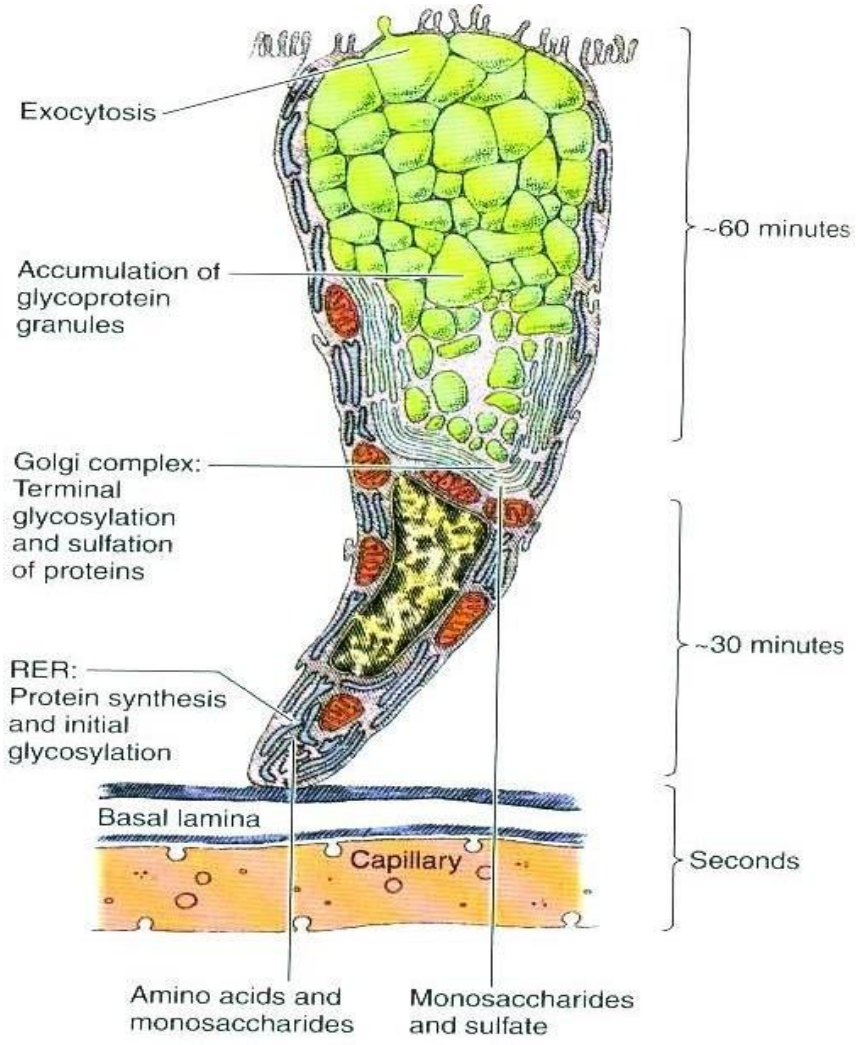
Contain about 300 cilia per cell.

Function :

Beating of their cilia moves the mucus upward



GOBLET CELLS



Flask shaped

Apical region - secretory vacoules (mucinogen granules) – lost during H & E preparation and cells appear empty.

basal region - nucleus, RER.

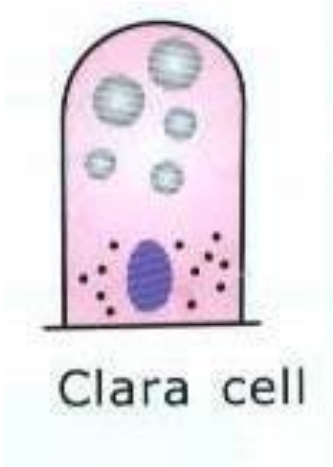
Function : produce mucus which traps the dust particles.

Location : trachea – tertiary bronchi

CLARA CELLS

Cuboidal cells , with microvilli on apical surface.

Location : terminal bronchioles.



Function :

Secrete lipoprotein, forms protective layer of surface active agent throughout lower respiratory tract , reduces surface tension in alveoli. Prevents adhesion between walls forming lumen of air passage especially during expiration.

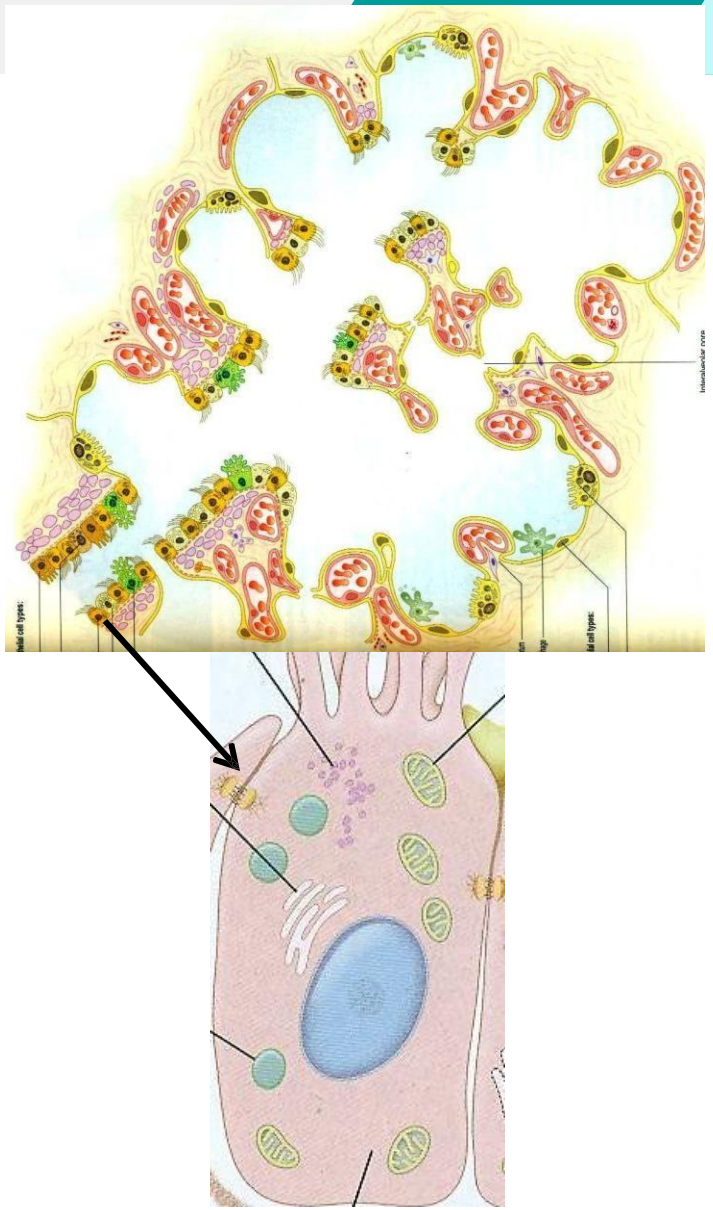
BRUSH CELLS

Columnar cells with microvilli on their free surfaces.

Location : conducting air passage.

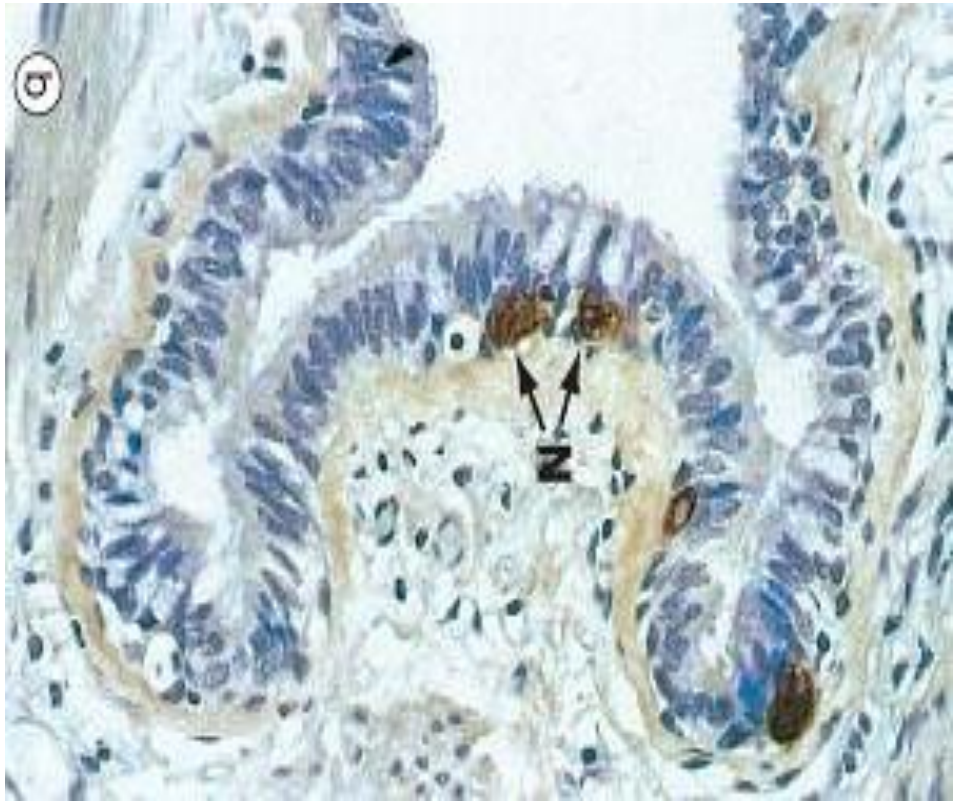
Function :

They are in contact with afferent nerve fibre basally so considered to be sensory in function



NEUROENDOCRINE CELL

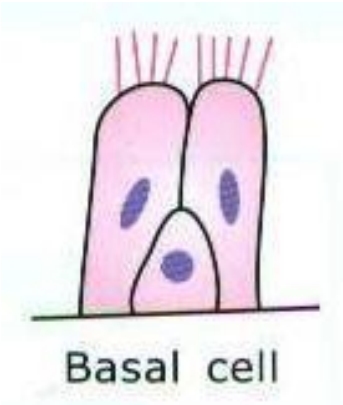
- Kulchitsky or ‘k’ cells.
- Contain electron dense secretory granules.
- Constitute part of diffuse endocrine system [APUD] cells.
- Found in basal part of epithelium
- Secrete serotonin ; calcitonin.
- Regulate local lung function – airway smooth muscle tone, mucus production, local blood flow



BASAL CELLS

Small pyramidal cells in contact with basal lamina.

Location : airway passage lined by psuedostratified epithelium.

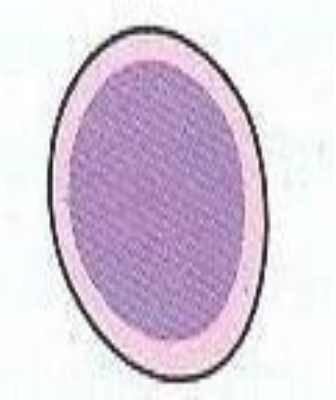


Function :

Act as mitotic stem cells for other type of epithelial cells

LYMPHOCYTES

- Small lymphocytes mainly T cells .
- Derived from MALT cells
- Location :conducting airway passage



Function :

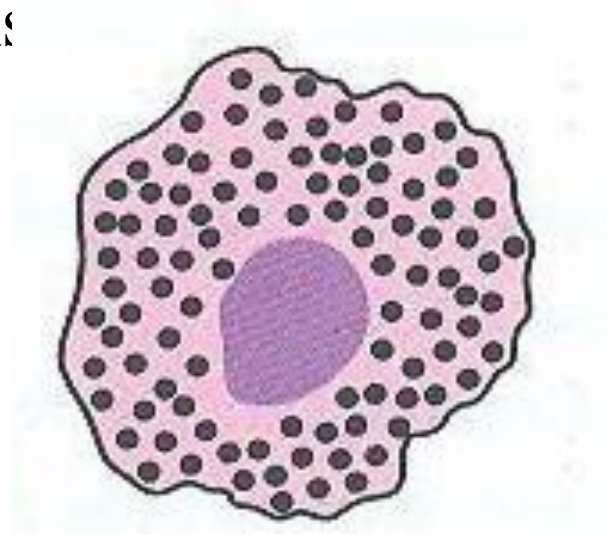
Protect the epithelial surface.

MAST CELL

Location : basal region of epithelium.

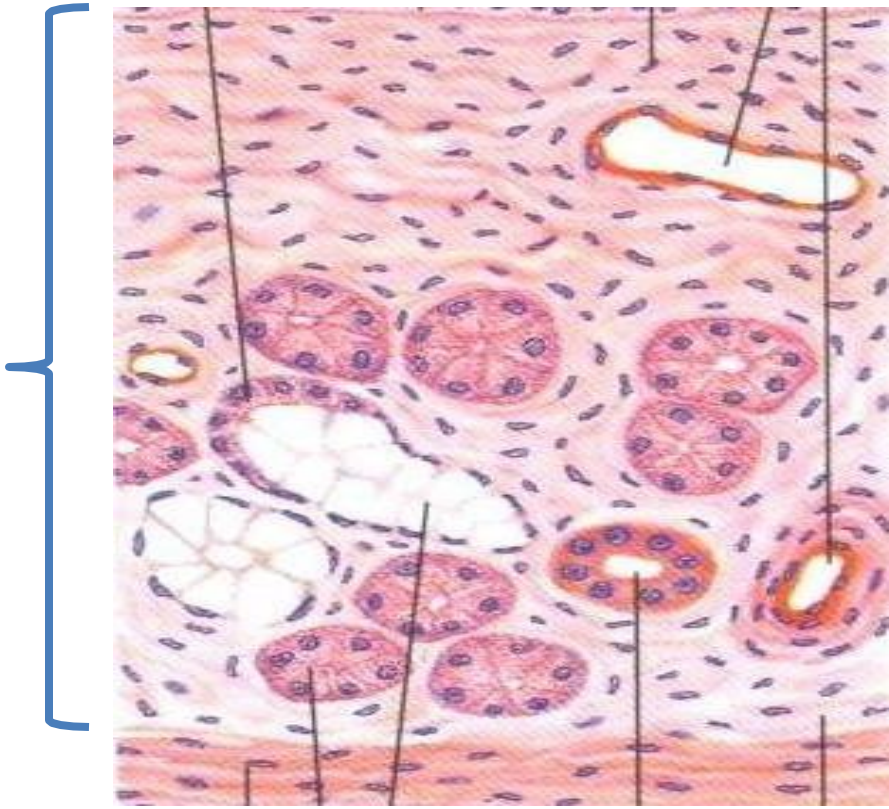
Function :

They release histamine containing granules in response to irritants, including inhaled allergens



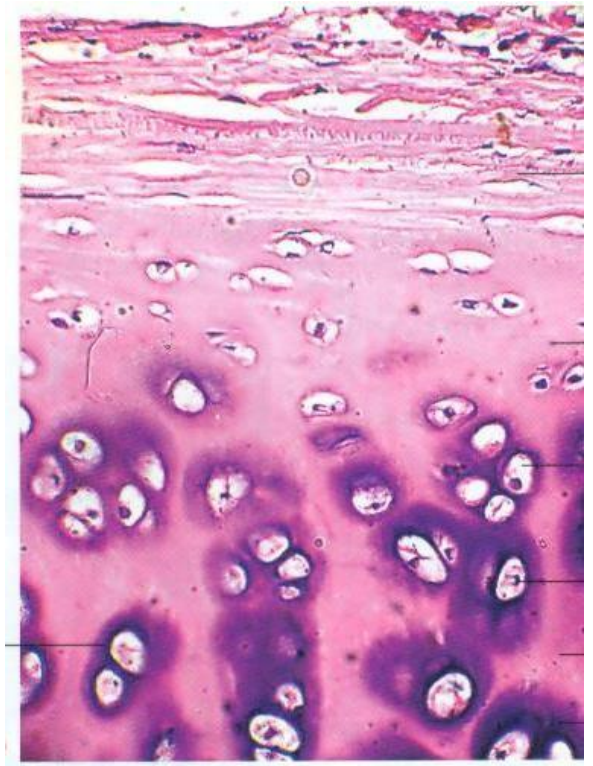
SUBMUCOSA

- It contains loose connective tissue .
- Mixed glands (seromucous).
- Duct of these glands open into suface epithelium.
- Contains fat cells , blood vessels and lymphatics



CARTILAGE LAYER

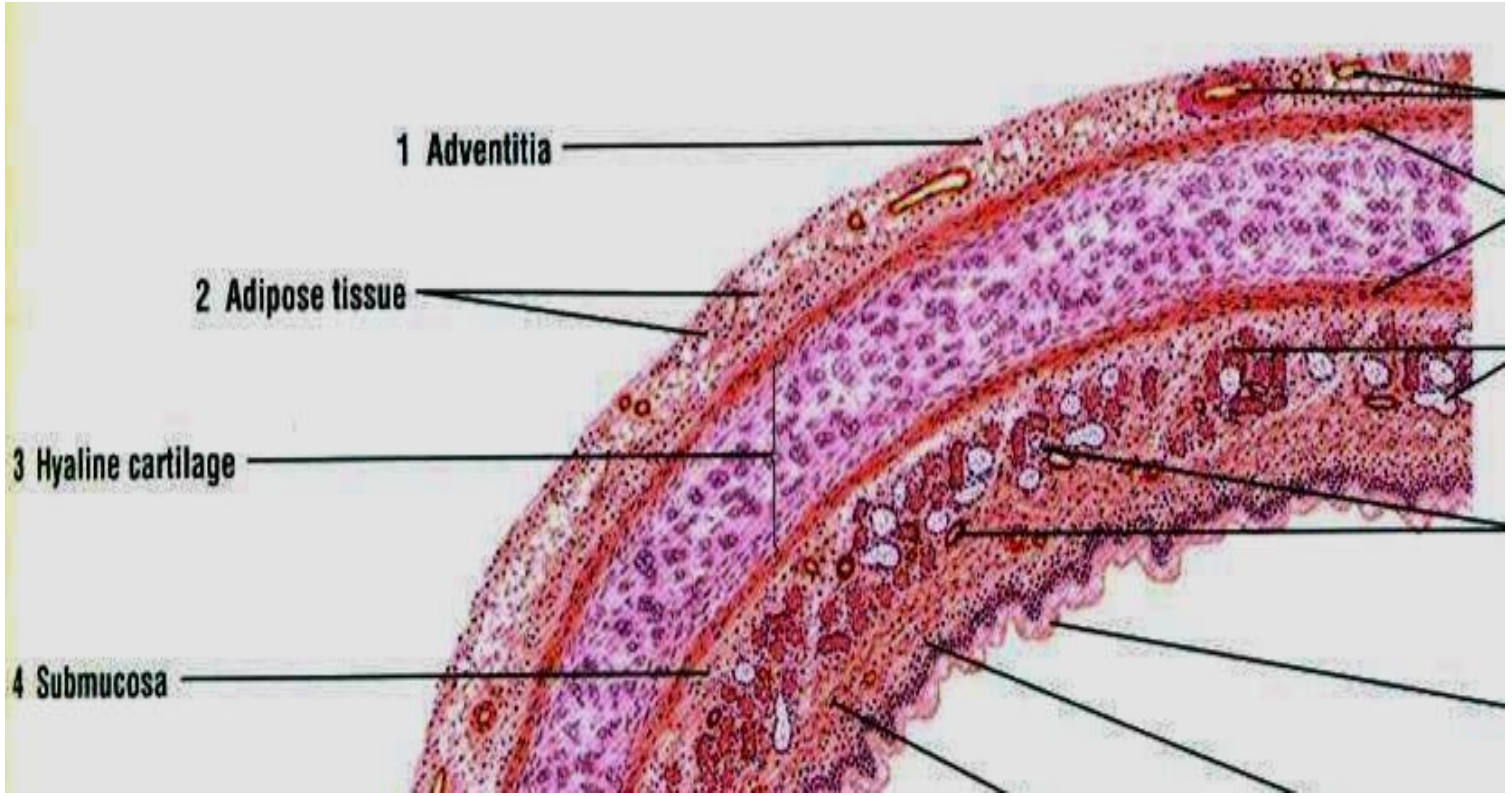
- Hyaline cartilage .
- Homogenous matrix , with collagen fibres and ground substance.
- Chondrocytes occupy lacunae in matrix [cell nest], synthesis of collagen fibres , ground substance.
- Perichondrium – two layers outer fibrous; inner cellular



- **Function**:-Provide flexibility to tracheal pipe
- Maintains patency of lumen

ADVENTIA

- It is layer of fibroelastic connective tissue.
- Contains blood vessels , nerves , adipose tissue.

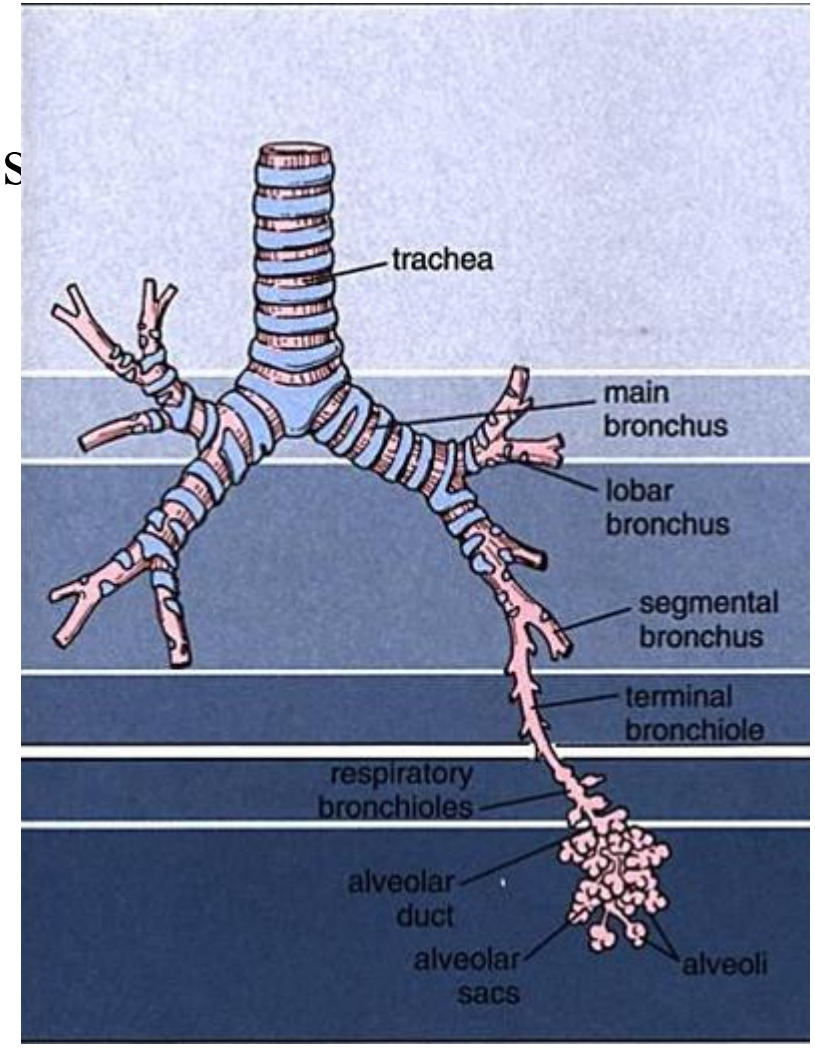


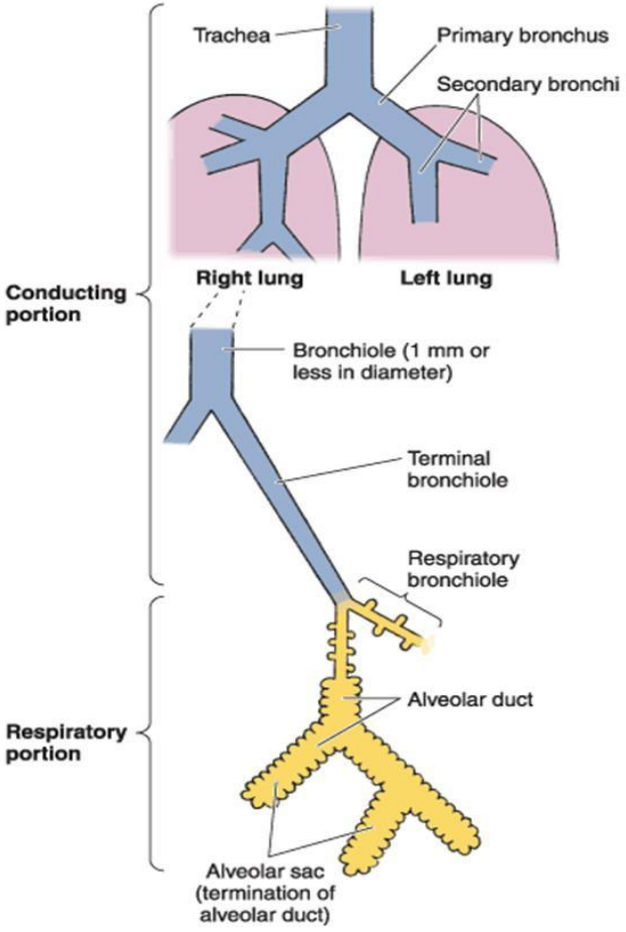
HISTOLOGY OF CONDUCTING SYSTEM

- Trachea
- The primary bronchus divides

into –secondary/lobar

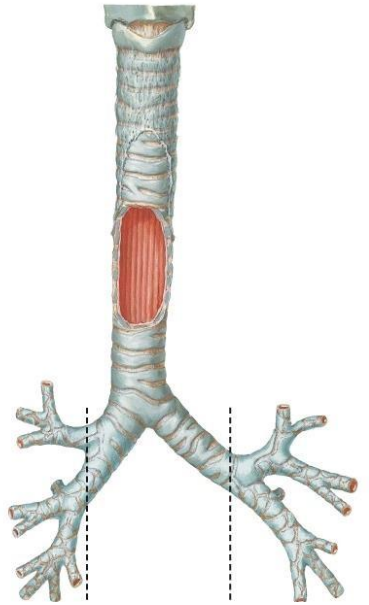
- Tertiary/segmental
- Bronchioles
- Terminal bronchiole.

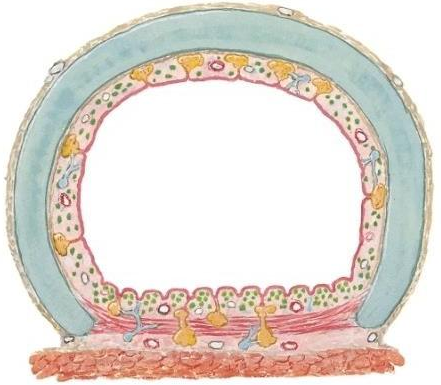




TRACHEA

- It is continuation of larynx.
- It begins at lower border of cricoid cartilage, opposite C6 vertebra.
- It divides – right and left principal bronchi opposite the sternal angle.
- Length:- 10 – 11cm.
- Structure:- in front 16-20 ‘C’ shaped cartilaginous rings , behind it’s deficient and completed by trachealis muscle.





HISTOLOGY OF TRACHEA

Mucosa:-

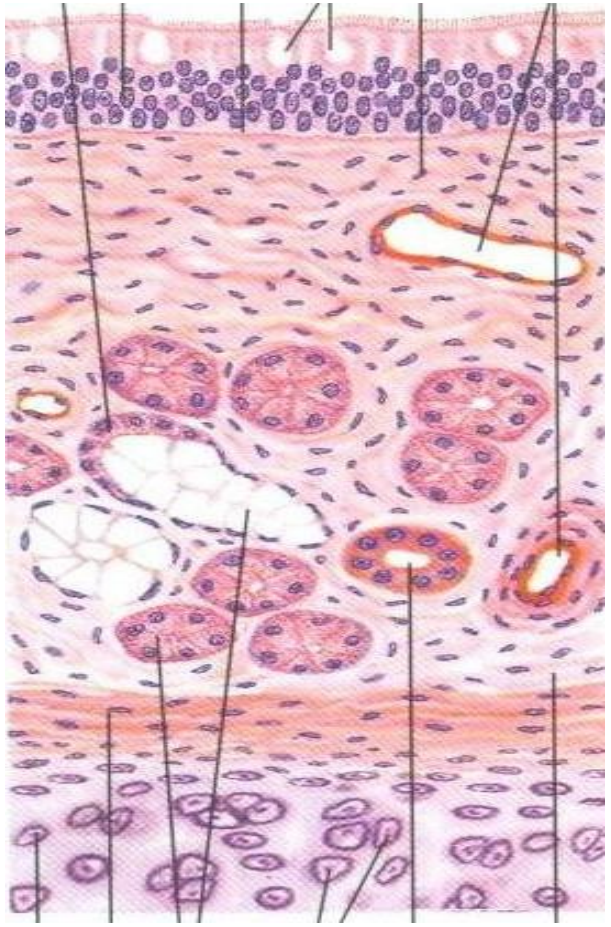
Epithelium-thick , pseudostratified ciliated columnar with goblet cells.

Lamina propria:-

Loose connective tissue rich in elastic fibres also contain lymphocytes and mast cells.

Submucosa:-

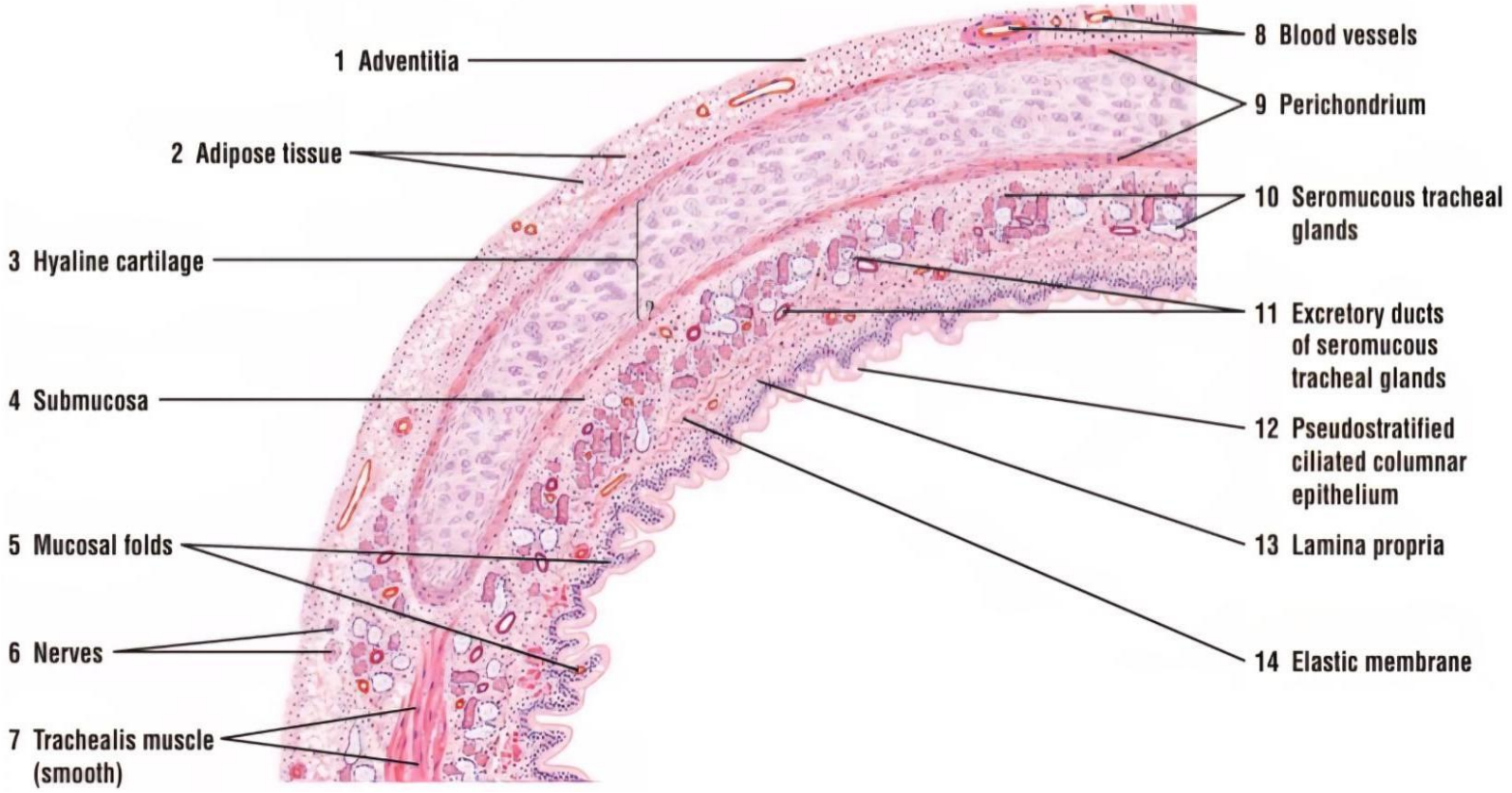
Loose connective tissue with seromucous glands.



HISTOLOGY OF TRACHEA

- **Cartilage layer**:-C shaped hyaline cartilagenous rings, posteriorly cartilage rings are bridged by smooth muscle [trachealis], fibroelastic connective tissue.

- **Adventia** :-layer of connective tissue rich in elastic fibers.



HISTOLOGY OF PRIMARY/PRINCIPAL BRONCHI

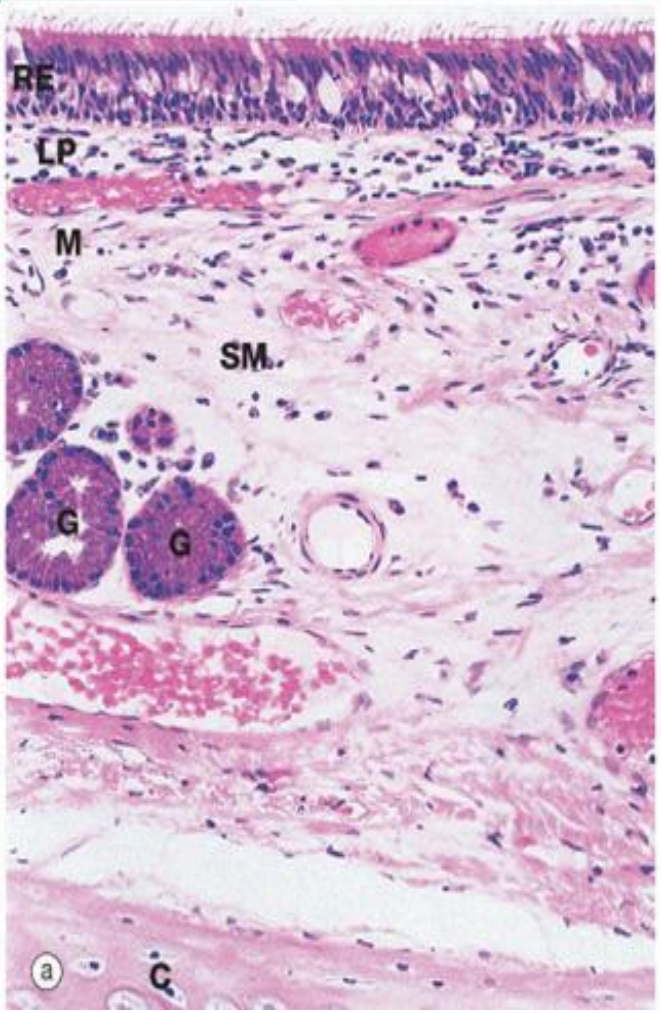
It is similar to trachea.

Epithelium less tall , contain fewer goblet cell.

Lamina propria : more dense , large quantity of elastin.

Submucosa : Separated from lamina propria by discontinuous layer of smooth muscle , seromucinous glands.

- Cartilage framework arranged into flattened interconnected plates.



HISTOLOGY OF SECONDARY/LOBAR BRONCHI

Mucosa

Epithelium : Psuedostratified ciliated columnar with few goblet cells.

Lamina propria : elastic fibers.

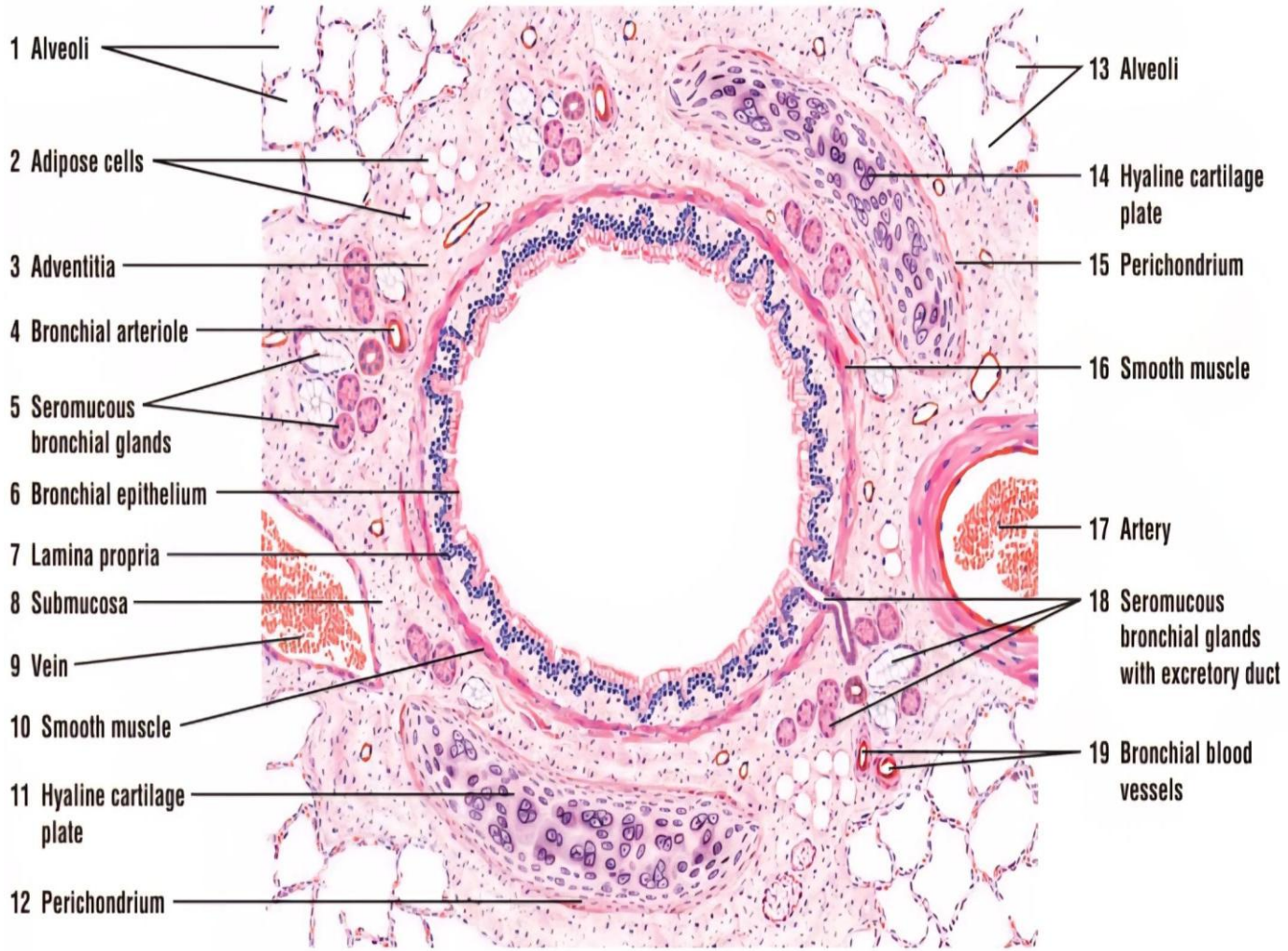
Mucosa is thrown into folds by contraction of smooth muscle.

Smooth muscle layer.

Submucosa : seromucous glands.

Cartilage layer : isolated plates of hyaline cartilage.

Adventia : connective tissue , blood vessels , adipose tissue .

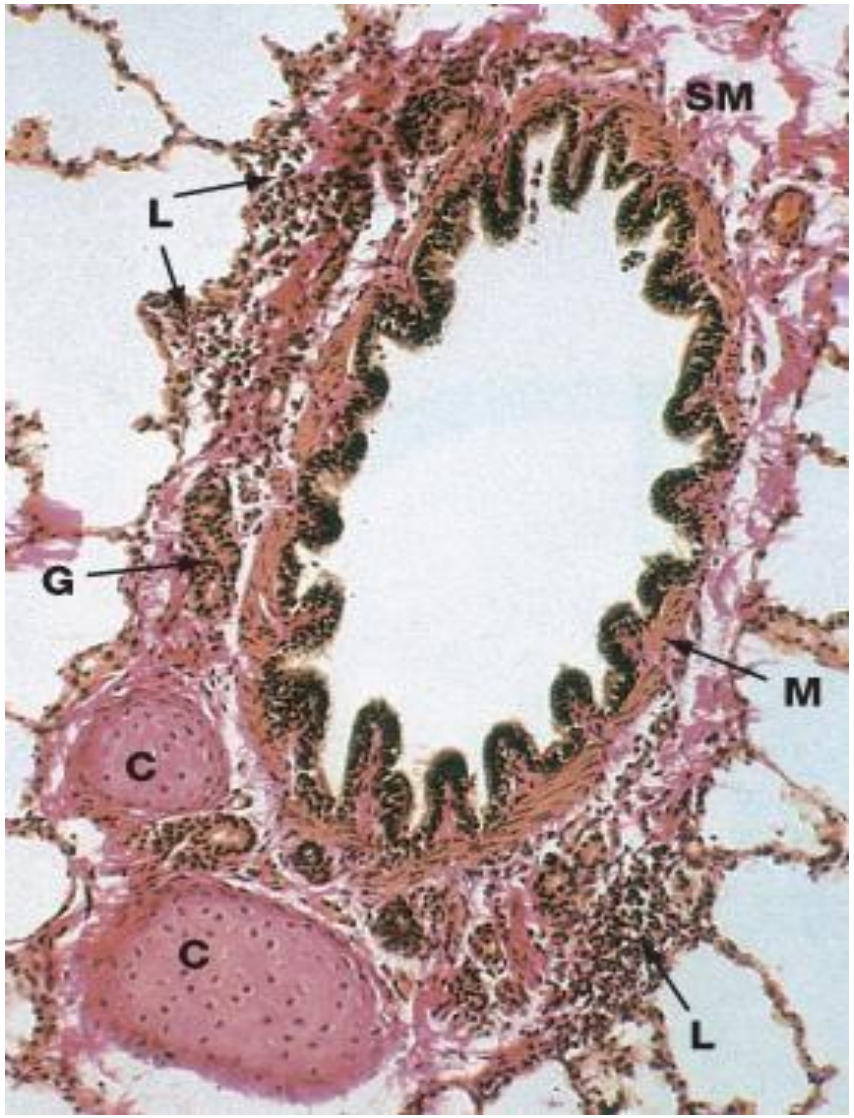


HISTOLOGY OF TERTIARY/SEGMENTAL BRONCHI

Mucosa :
epithelium – tall columnar with little psuedostratification , less goblet cells. **Lamina propria** :
Elastic fibres , smooth muscle spirally arranged.
Submucosa :
Sparse seromucinous gland.
Cartilage layer :
Few thin irregular plates.
Adventia :
Aggregations of lymphocytes [BALT].

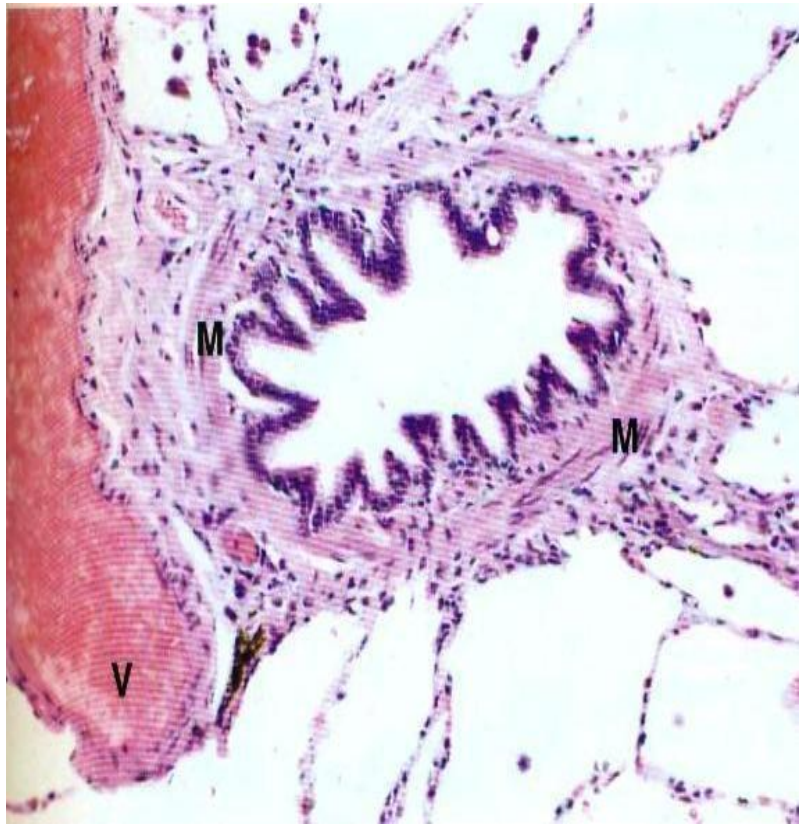
It is component of diffuse mucosa associated lymphoid tissue.

Function :Provide protection to mucosa and submucosa.



BRONCHIOLE

Mucosa :
Epithelium : simple ciliated columnar with goblet cells.
Lamina propria :
Connective tissue with elastic fibers.
Smooth muscle layers folds into lumen.
Submucosa :
Connective tissue , no glands.
No cartilage plates.
Adventia :
Connective tissue with lymphatics and blood vessels.



TERMINAL BRONCHIOLE

Mucosa :

Epithelium :simple cuboidal ciliated with clara cells.

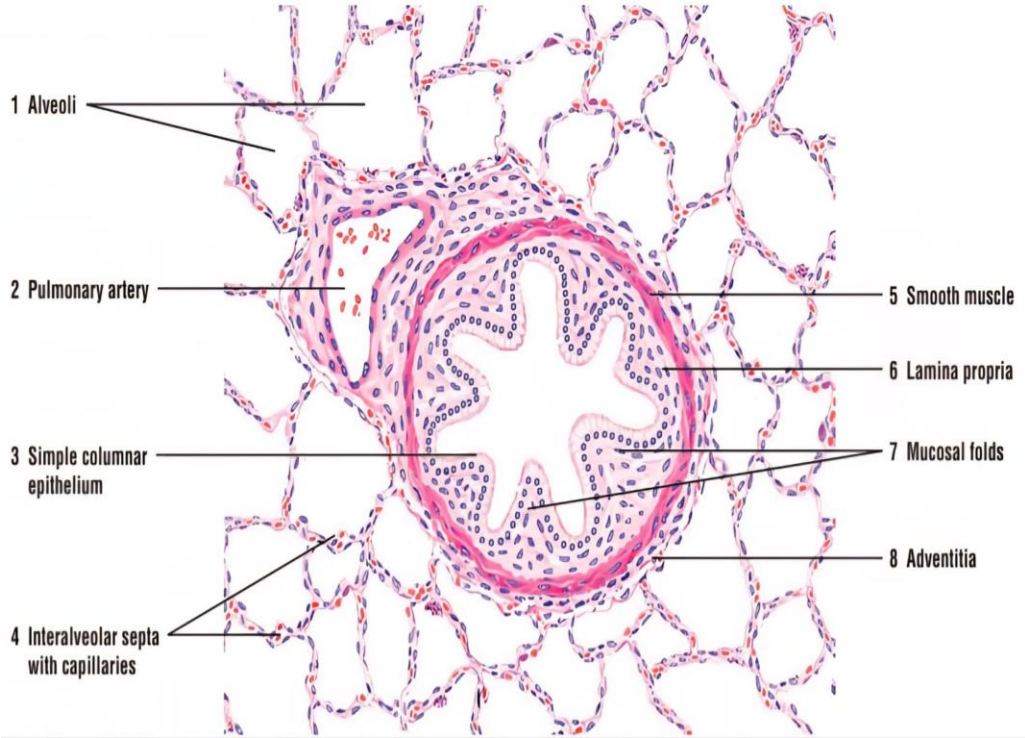
Goblet cells are absent.

Lamina propria :

Connective tissue with smooth muscle fibers.

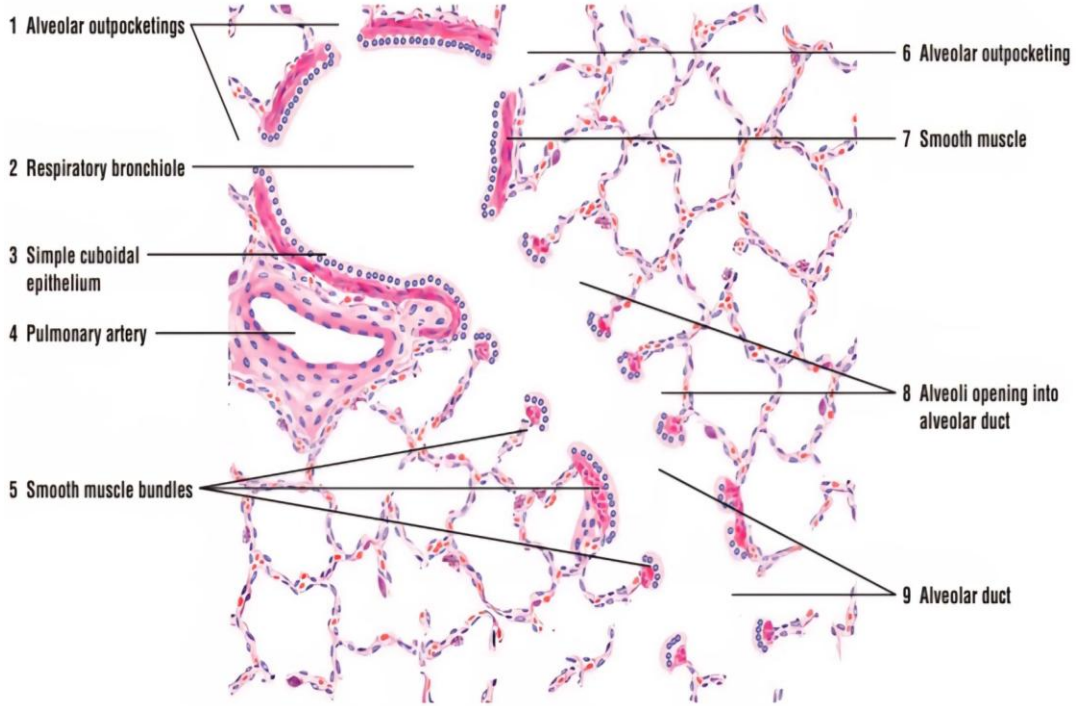
Submucosa : No glands. No cartilage plates.

Adventia : Connective tissue , blood vessels.



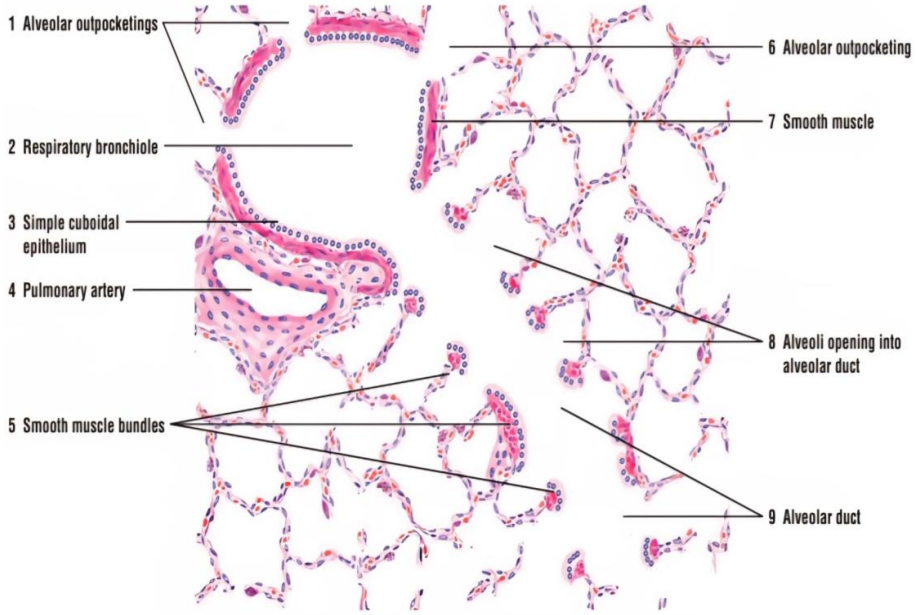
RESPIRATORY BRONCHIOLE

- Constitute transitional zone.
- Involved in both air condition ,gaseous exchange.
- Epithelium : low columnar/cuboidal/ciliated, with clara cells.
- Lamina propria : connective tissue, smooth , elastic fibers
- Single alveolar out pocketing present in walls.



ALVEOLAR DUCTS

- Arise from respiratory bronchiole
- Epithelium : squamous epithelial cells,supported by smooth muscle,fibroelastic connective tissue.
- Alveoli opens along length of duct,in section appear knobs of smooth muscle between adjacent alveoli.
- It opens into atria alveolar sac and alveoli.



ALVEOLI

- Sac like evagination present at terminal part of bronchial tree.
- Site of gas exchange between air and blood.
- In section – honey comb appearance.
- In between two alveoli there is interalveolar septum.

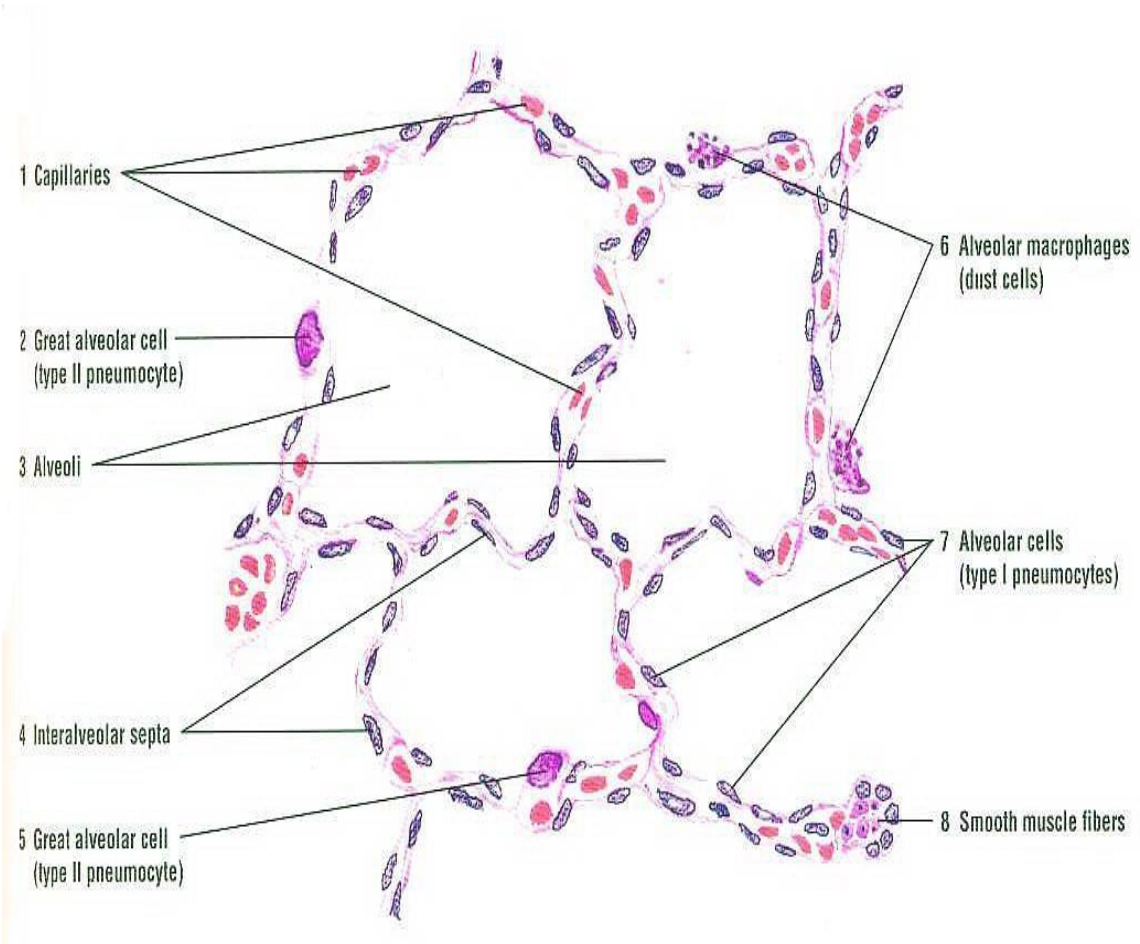


ALVEOLI

- Interalveolar septa contain network of capillaries, supported by reticular and elastic fibers.

Epithelium :

- Simple squamous alveolar cells

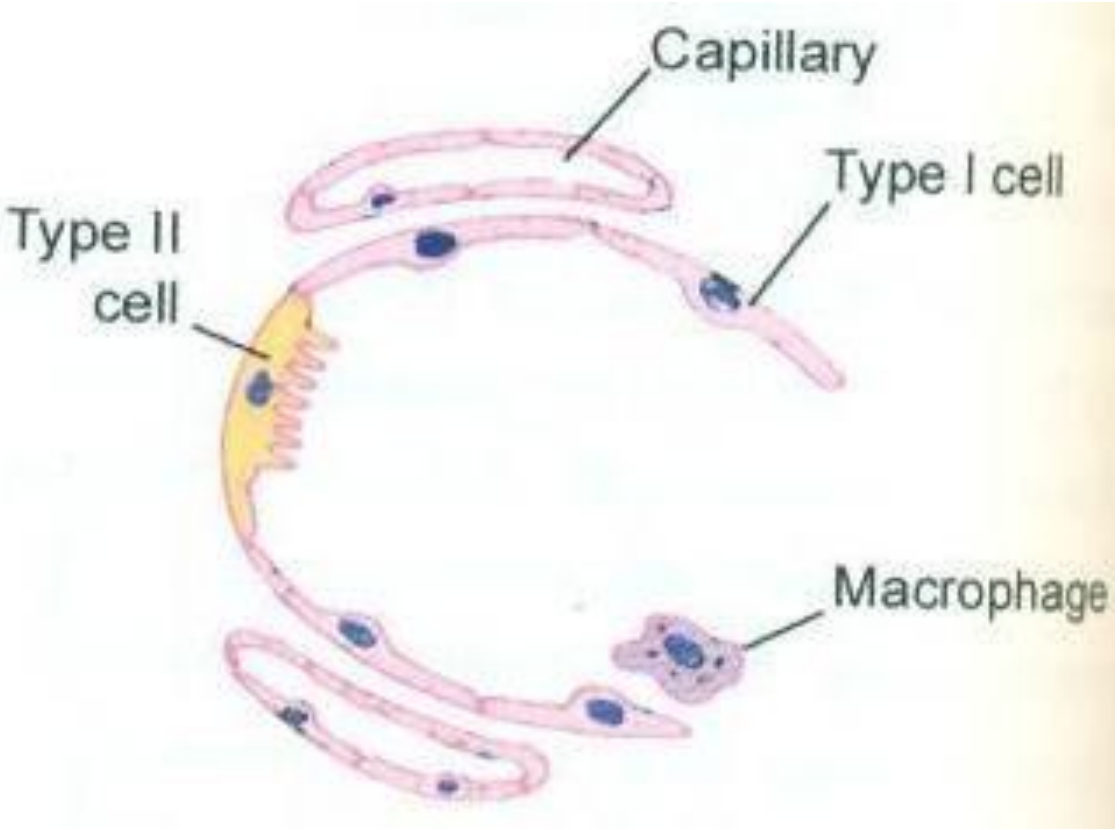


ALVEOLAR CELLS

Type 1 alveolar cells.

Type 2 alveolar cells.

Alveolar macrophage [dust cell].



TYPE 1 ALVEOLAR CELLS / PNEUMOCYTES

Lined by thin squamous cells

Cells are joined to one another by occluding junction.

Cover 97% of alveolar surface.

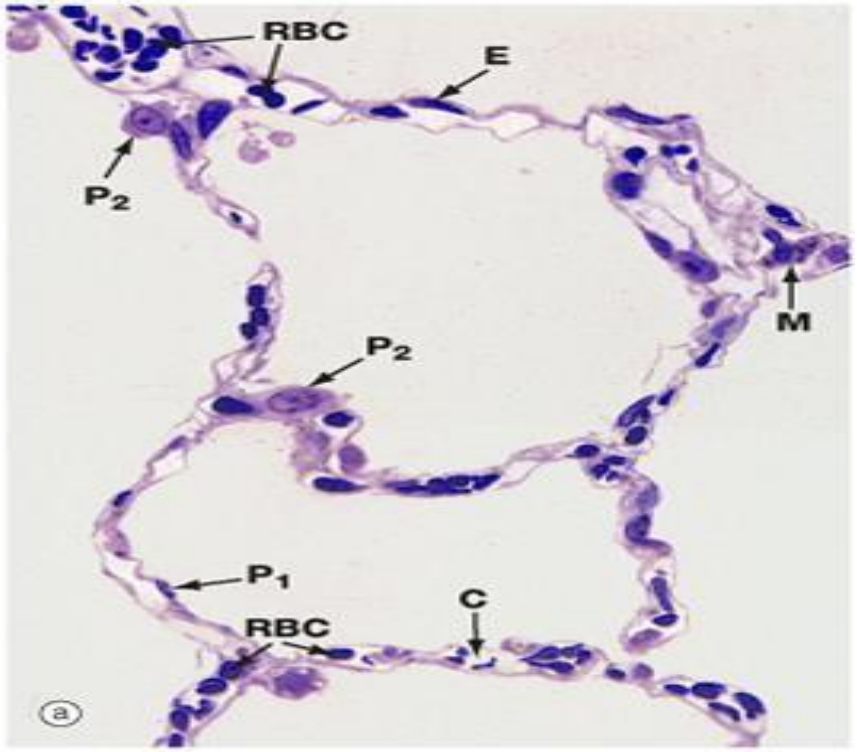
Contain abundant pinocytic vesicles.

Function:

Absorption of surfactant.

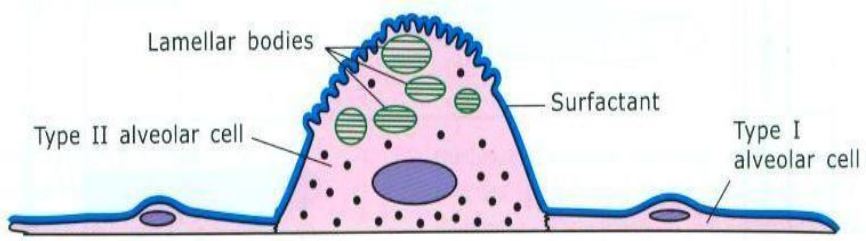
Removal of particulate contaminants.

Blood – air barrier.



TYPE 2 ALVEOLAR CELLS

- Great alveolar cells/septal/type 2 pneumocyte.
- Cuboidal cell , microvilli.
- Covers 3-5% of alveolar surface.



- Foamy vacuolated cytoplasm –lamellated bodies, which consists of concentric whorls of phospholipids, precursor of alveolar surfactant.

ALVEOLAR SURFACTANT

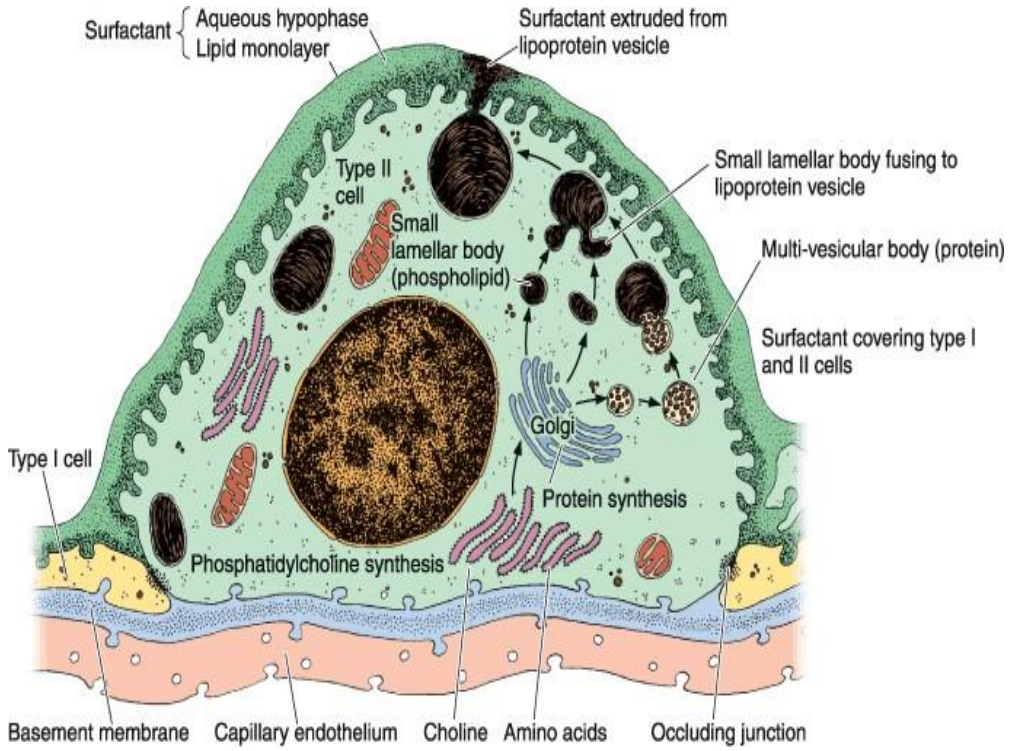
Lamellated bodies.

Present over alveolar surface.

Functions :

Lowers surface tension.

Bactericidal activity.



Modulates immune responses to virus bacteria and fungi.

ALVEOLAR MACROPHAGE

Derived from monocytes

Found in interalveolar connective tissue septa.

Cytoplasm –phagocytosed carbon and dust particles.

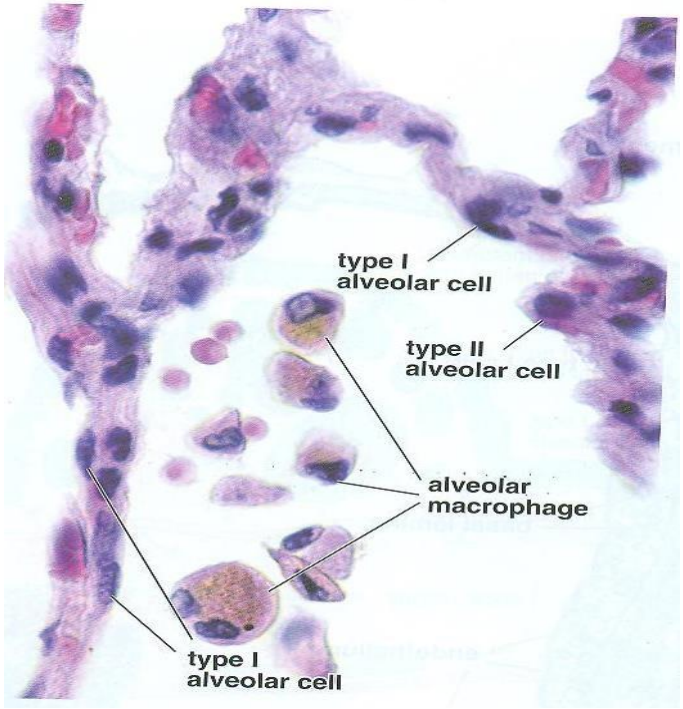
Function :

Phagocytose dust particles – dust cells.

Engulf carbon and tar-smokers.

Phagocytose bacteria.

Heart failure cells.



BLOOD-AIR BARRIER

Separate alveolar air from blood.

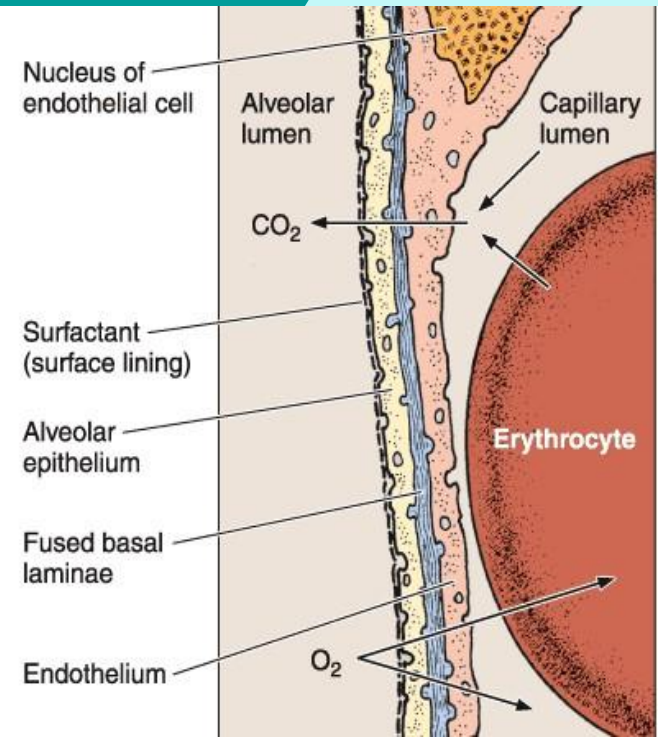
Components :

Non fenestrated endothelium of capillary.

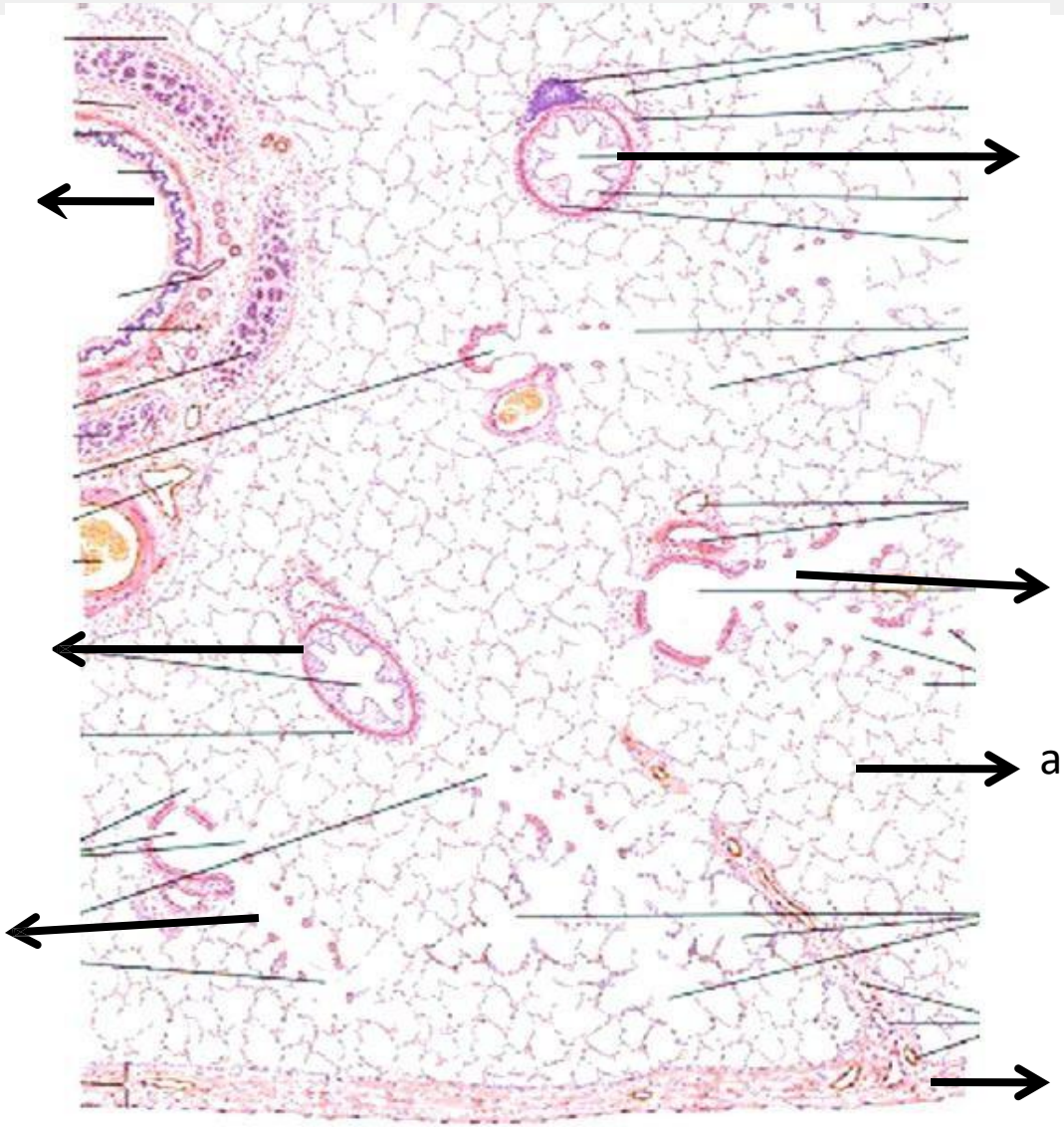
Squamous epithelium lining of alveolus.

Fused basal lamina of endothelium and epithelium.

Thin blood-air barrier facilitates diffusion of O2 and CO2



HISTOLOGY OF LUNGS



THANK YOU